

EXTENSION 2

Stage 6

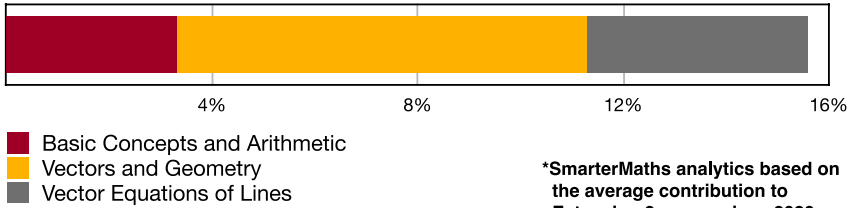
Vectors (Ext2), V1 Working With Vectors (Ext2)

Basic Concepts and Arithmetic

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Exam Equivalent Time: 117 minutes (based on allocation of 1.5 minutes per mark)

V1 Working with Vectors



*SmarterMaths analytics based on the average contribution to Extension 2 exams since 2020.

HISTORICAL CONTRIBUTION

- V1 Working With Vectors has contributed an impressive 15.5% per new syllabus Ext2 exam since it was introduced in 2020.
- We have split this topic into three categories for analysis purposes which are: 1-Basic Concepts and Arithmetic (3.3%), 2-Vectors and Geometry (8.0%) and 3-Vectors and Vector Equations of Lines (4.2%).
- This analysis looks at Basic Concepts and Arithmetic.

HSC ANALYSIS - What to expect and common pitfalls

- Basic Concepts and Arithmetic (3.3%) has been examined in longer answer questions in every new syllabus Ext2 exam to date, including an additional multiple choice question in 2020.
- All questions on this topic area have been examiner softballs in the band 2–3 difficulty range, with each longer answer question worth a solid 3-marks.
- Angles between vectors has been the most commonly examined topic, examined and well answered in each year 2020-2023, including perpendicular vectors in 2020.
- Dedicated questions on the more difficult vector projection concept are yet to appear in new syllabus exams with the exception of 2021 Ext2 10 MC which combined complex numbers with this concept and caused problems (mean mark 30%).

Questions

1. Vectors, EXT2 V1 2020 HSC 1 MC

What is the length of the vector $-\tilde{i} + 18\tilde{j} - 6\tilde{k}$?

- 5
- 19
- 25
- 361

2. Vectors, EXT2 V1 2011 SPEC2 12 MC

The angle between the vectors $3\tilde{i} + 6\tilde{j} - 2\tilde{k}$ and $2\tilde{i} - 2\tilde{j} + \tilde{k}$, correct to the nearest tenth of a degree, is

- A. 2.0°
- B. 91.0°
- C. 112.4°
- D. 121.3°

3. Vectors, EXT2 V1 2012 SPEC2 16 MC

The distance between the points $P(-2, 4, 3)$ and $Q(1, -2, 1)$ is

- A. 7
- B. $\sqrt{21}$
- C. $\sqrt{31}$
- D. 49

4. Vectors, EXT2 V1 2013 SPEC2 14 MC

The distance from the origin to the point $P(7, -1, 5\sqrt{2})$ is

- A. $7\sqrt{2}$
- B. 10
- C. $6 + 5\sqrt{2}$
- D. 100

5. Vectors, EXT2 V1 2020 SPEC2 16 MC

Let $\vec{a} = \vec{i} + 2\vec{j} + 2\vec{k}$ and $\vec{b} = 2\vec{i} - 4\vec{j} + 4\vec{k}$, where the acute angle between these vectors is θ .

The value of $\sin(2\theta)$ is

- A. $\frac{1}{9}$
- B. $\frac{4\sqrt{5}}{9}$
- C. $\frac{4\sqrt{5}}{81}$
- D. $\frac{8\sqrt{5}}{81}$

6. Complex Numbers, EXT2 N2 2021 HSC 10 MC

Consider the two non-zero complex numbers z and w as vectors.

Which of the following expressions is the projection of z onto w ?

- A. $\frac{\operatorname{Re}(zw)}{|w|}w$
- B. $\left|\frac{z}{w}\right|w$
- C. $\operatorname{Re}\left(\frac{z}{w}\right)w$
- D. $\frac{\operatorname{Re}(z)}{|w|}w$

7. Vectors, EXT2 V1 2023 HSC 10 MC

Consider any three-dimensional vectors $\vec{a} = \vec{OA}$, $\vec{b} = \vec{OB}$ and $\vec{c} = \vec{OC}$ that satisfy these three conditions

- $\vec{a} \cdot \vec{b} = 1$
- $\vec{b} \cdot \vec{c} = 2$
- $\vec{c} \cdot \vec{a} = 3$.

Which of the following statements about the vectors is true?

- A. Two of $\vec{a}$, $\vec{b}$ and $\vec{c}$ could be unit vectors.
- B. The points A , B and C could lie on a sphere centred at O .
- C. For any three-dimensional vector $\vec{a}$, vectors $\vec{b}$ and $\vec{c}$ can be found so that $\vec{a}$, $\vec{b}$ and $\vec{c}$ satisfy these three conditions.
- D. $\forall \vec{a}, \vec{b}$ and $\vec{c}$ satisfying the conditions, $\exists \vec{r}, \vec{s}$ and $\vec{t}$ such that $\vec{r}, \vec{s}$ and $\vec{t}$ are positive real numbers and $\vec{r}\vec{a} + \vec{s}\vec{b} + \vec{t}\vec{c} = \vec{0}$.

8. Vectors, EXT2 V1 EQ-Bank 2

Find the angle between the vectors $\vec{r} = \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$ and $\vec{s} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$, giving the angle in degrees correct to 1 decimal place. (3 marks)

9. Vectors, EXT2 V1 2021 HSC 11c

Find the angle between the vectors $\vec{a} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}$, giving the angle in degrees correct to 1 decimal place. (3 marks)

10. Vectors, EXT2 V1 2023 HSC 11b

Find the angle between the vectors

$\vec{a} = \vec{i} + 2\vec{j} - 3\vec{k}$
 $\vec{b} = -\vec{i} + 4\vec{j} + 2\vec{k}$

giving your answer to the nearest degree. (3 marks)

11. Vectors, EXT2 V1 EQ-Bank 1

Prove that the vectors $4\vec{i} + 5\vec{j} - 2\vec{k}$ and $-5\vec{i} + 6\vec{j} + 5\vec{k}$ are perpendicular. (2 marks)

12. Vectors, EXT2 V1 EQ-Bank 3

If $\vec{a} = 3\vec{i} - \vec{j}$ and $\vec{b} = -2\vec{i} + 6\vec{j} + 2\vec{k}$

- i. Calculate $\vec{a} - \frac{1}{2}\vec{b}$ (2 marks)
- ii. Find $\hat{\vec{b}}$ (2 marks)

13. Vectors, EXT2 V1 SM-Bank 5

Find the value(s) of m so that the vectors $\vec{a} = 2\vec{i} + m\vec{j} - 3\vec{k}$ and $\vec{b} = m^2\vec{i} - \vec{j} + \vec{k}$ are perpendicular. (2 marks)

14. Vectors, EXT2 V1 SM-Bank 12

Two vectors are given by $\vec{a} = 4\vec{i} + m\vec{j} - 3\vec{k}$ and $\vec{b} = -2\vec{i} + n\vec{j} - \vec{k}$, where $m, n \in \mathbb{R}^+$.

If $|\vec{a}| = 10$ and $\vec{a}$ is perpendicular to $\vec{b}$, then find m and n . (2 marks)

15. Vectors, EXT2 V1 2020 HSC 11d

Consider the two vectors $\vec{u} = 2\vec{i} - \vec{j} + 3\vec{k}$ and $\vec{v} = p\vec{i} + \vec{j} + 2\vec{k}$.

For what values of p are $\vec{u} - \vec{v}$ and $\vec{u} + \vec{v}$ perpendicular? (3 marks)

16. Vectors, EXT2 V1 2022 HSC 11d

A triangle is formed in three-dimensional space with vertices $A(1, -1, 2)$, $B(0, 2, -1)$ and $C(2, 1, 1)$.

Find the size of $\angle ABC$, giving your answer to the nearest degree. (3 marks)

17. Vectors, EXT2 V1 2022 SPEC1 6

Find the cosine of the acute angle between the vectors $\vec{a} = 2\vec{i} - 3\vec{j} + 6\vec{k}$ and $\vec{b} = \vec{i} + 2\vec{j} + 2\vec{k}$. (2 marks)

18. Vectors, EXT2 V1 EQ-Bank 12

Two planes are flying in directions given by the vectors $-200\vec{i} + 400\vec{j} - 3\vec{k}$ and $300\vec{i} + 250\vec{j} + \vec{k}$.

A person in the flight control centre is plotting their paths on a map.

Calculate the acute angle between their projected flight paths, giving your answer correct to one decimal place. (2 marks)

19. Vectors, EXT2 V1 EQ-Bank 6

Use vector methods to find the coordinates of the point that divides the interval joining the points $A(-1, 3, 2)$ and $B(7, -1, -6)$ in the ratio $1:3$. (3 marks)

20. Vectors, EXT2 V1 EQ-Bank 9

A parallelogram is formed by joining the points $P(-2, 1, 4)$, $Q(1, 4, 5)$, $R(0, 2, 3)$ and $S(a, b, c)$.

Use vector methods to find a, b and c . (2 marks)

21. Vectors, EXT2 V1 SM-Bank 11

Given $\lambda_1\vec{a} + \lambda_2\vec{b} = \begin{bmatrix} 50 \\ -45 \\ -8 \end{bmatrix}$, find λ_1 and λ_2 if

$$\vec{a} = \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} 3 \\ -2 \\ -3 \end{bmatrix}. \text{ (2 marks)}$$

22. Vectors, EXT2 V1 SM-Bank 4

Consider the three vectors

$$\vec{a} = \vec{i} - \vec{j} + 2\vec{k}, \vec{b} = \vec{i} + 2\vec{j} + m\vec{k} \text{ and } \vec{c} = \vec{i} + \vec{j} - \vec{k}, \text{ where } m \in \mathbb{R}.$$

i. Find the value(s) of m for which $\left| \vec{b} \right| = 2\sqrt{3}$. (2 marks)

ii. Find the value of m such that $\vec{a}$ is perpendicular to $\vec{b}$. (1 mark)

23. Vectors, EXT2 V1 2013 SPEC1 3

The coordinates of three points are $A \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$, $B \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix}$ and $C \begin{pmatrix} 3 \\ 5 \\ 2 \end{pmatrix}$.

i. Find $\overrightarrow{AB}$. (1 mark)

ii. The points A, B and C are the vertices of a triangle.

Prove that the triangle has a right angle at A . (2 marks)

iii. Find the length of the hypotenuse of the triangle. (1 mark)

24. Vectors, EXT2 V1 2014 SPEC1 1

Consider the vector $\vec{a} = \sqrt{3}\vec{i} - \vec{j} - \sqrt{2}\vec{k}$, where $\vec{i}, \vec{j}$ and $\vec{k}$ are unit vectors in the positive directions of the x, y and z axes respectively.

i. Find the unit vector in the direction of $\vec{a}$. (1 mark)

ii. Find the acute angle that $\vec{a}$ makes with the positive direction of the x -axis. (2 marks)

iii. The vector $\vec{b} = 2\sqrt{3}\vec{i} + m\vec{j} - 5\vec{k}$.

Given that $\vec{b}$ is perpendicular to $\vec{a}$, find the value of m . (2 marks)

25. Vectors, EXT2 V1 EQ-Bank 4

Find all possible values of b given $\left| \overrightarrow{PQ} \right| = \sqrt{76}$ where $P(-5, b, -2)$ and $Q(-3, -2, 4)$. (3 marks)

26. Vectors, EXT2 V1 2017 SPEC1 10

Consider the vectors $\vec{a} = -\vec{i} - 2\vec{j} + 3\vec{k}$ and $\vec{b} = 2\vec{i} + c\vec{j} + \vec{k}$.

Find the value of c , $c \in \mathbb{R}$, if the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$. (4 marks)

27. Vectors, EXT2 V1 SM-Bank 25

The acute angle θ is the angle between the vectors $\vec{a} = -2\vec{i} + 2\vec{j} - \vec{k}$ and $\vec{b} = -4\vec{i} + 4\vec{j} + 7\vec{k}$.

Find the exact value of $\sin(2\theta)$. (2 marks)

28. Vectors, EXT2 V1 SM-Bank 7

If θ is the angle between $\vec{a} = \sqrt{3}\vec{i} + 4\vec{j} - \vec{k}$ and $\vec{b} = \vec{i} - 4\vec{j} + \sqrt{3}\vec{k}$, then find $\cos(2\theta)$. (2 marks)

29. Vectors, EXT2 V1 SM-Bank 8

If $\vec{a} = -2\vec{i} - \vec{j} + 3\vec{k}$ and $\vec{b} = -m\vec{i} + \vec{j} + 2\vec{k}$, where m is a real constant, find m such that the vector $\vec{a} - \vec{b}$ will be perpendicular to vector $\vec{b}$. (2 marks)

30. Vectors, EXT2 V1 2020 SPEC1 5

Let $\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$ and $\vec{b} = \vec{i} + m\vec{j} - \vec{k}$, where m is an integer.

The projection of $\vec{a}$ onto $\vec{b}$ is $-\frac{11}{18}(\vec{i} + m\vec{j} - \vec{k})$.

a. Find the value of m . (3 marks)

b. Find the component of $\vec{a}$ that is perpendicular to $\vec{b}$. (1 mark)

31. Vectors, EXT2 V1 2017 SPEC1 5

Relative to a fixed origin, the points B , C and D are defined respectively by the position vectors $\vec{b} = \vec{i} - \vec{j} + 2\vec{k}$, $\vec{c} = 2\vec{i} - \vec{j} + \vec{k}$ and $\vec{d} = a\vec{i} - 2\vec{j}$ where a is a real constant.

Given that the magnitude of angle BCD is $\frac{\pi}{3}$, find a . (4 marks)

32. Vectors, EXT2 V1 SM-Bank 9

Points A , B and C have position vectors $\vec{a} = 2\vec{i} + \vec{j}$, $\vec{b} = 3\vec{i} - \vec{j} + \vec{k}$ and $\vec{c} = -3\vec{j} + \vec{k}$ respectively.

Find the cosine of angle ABC . (2 marks)

Worked Solutions

1. Vectors, EXT2 V1 2020 HSC 1 MC

$$\begin{aligned}\text{Length} &= \left| -\vec{i} + 18\vec{j} - 6\vec{k} \right| \\ &= \sqrt{(-1)^2 + 18^2 + (-6)^2} \\ &= \sqrt{361} \\ &= 19\end{aligned}$$

2. Vectors, EXT2 V1 2011 SPEC2 12 MC

$$\begin{aligned}\left| 3\vec{i} + 6\vec{j} - 2\vec{k} \right| &= \sqrt{9 + 36 + 4} = \sqrt{49} = 7 \\ \left| 2\vec{i} - 2\vec{j} + \vec{k} \right| &= \sqrt{4 + 4 + 1} = \sqrt{9} = 3 \\ \left(3\vec{i} + 6\vec{j} - 2\vec{k} \right) \cdot \left(2\vec{i} - 2\vec{j} + \vec{k} \right) &= 3 \times 2 + 6 \times (-2) + (-2) \times 1 \\ &= 6 - 12 - 2 \\ &= -8\end{aligned}$$

$$\begin{aligned}\cos\theta &= \frac{(3\vec{i} + 6\vec{j} - 2\vec{k}) \cdot (2\vec{i} - 2\vec{j} + \vec{k})}{\left| 3\vec{i} + 6\vec{j} - 2\vec{k} \right| \left| 2\vec{i} - 2\vec{j} + \vec{k} \right|} \\ &= -\frac{8}{21} \\ \therefore \theta &= \cos^{-1}\left(-\frac{8}{12}\right) \\ &\approx 112.4^\circ\end{aligned}$$

$$\Rightarrow C$$

3. Vectors, EXT2 V1 2012 SPEC2 16 MC

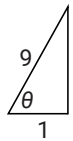
$$\begin{aligned}d &= \sqrt{(-2-1)^2 + (4-(-2))^2 + (3-1)^2} \\ &= \sqrt{9 + 36 + 4} \\ &= 7 \\ &\Rightarrow A\end{aligned}$$

4. Vectors, EXT2 V1 2013 SPEC2 14 MC

$$\begin{aligned} d &= \sqrt{(7-0)^2 + (-1-0)^2 + (5\sqrt{2}-0)^2} \\ &= \sqrt{49 + 1 + 25 \times 2} \\ &= 10 \\ &\Rightarrow B \end{aligned}$$

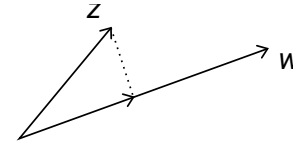
5. Vectors, EXT2 V1 2020 SPEC2 16 MC

$$\begin{aligned} \tilde{a} &= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, |\tilde{a}| = \sqrt{1+4+4} = 3 \\ \tilde{b} &= \begin{pmatrix} 2 \\ -4 \\ 4 \end{pmatrix}, |\tilde{b}| = \sqrt{4+16+16} = 6 \\ \tilde{a} \cdot \tilde{b} &= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -4 \\ 4 \end{pmatrix} = 2-8+8 = 2 \\ \cos\theta &= \frac{\tilde{a} \cdot \tilde{b}}{|\tilde{a}||\tilde{b}|} = \frac{2}{3 \times 6} = \frac{1}{9} \end{aligned}$$



$$\begin{aligned} \sin\theta &= \frac{\sqrt{81-1}}{9} = \frac{4\sqrt{5}}{9} \\ \therefore \sin(2\theta) &= 2 \times \frac{1}{9} \times \frac{4\sqrt{5}}{9} \\ &= \frac{8\sqrt{5}}{81} \\ &\Rightarrow D \end{aligned}$$

6. Complex Numbers, EXT2 N2 2021 HSC 10 MC



$$\text{Let } z = a + ib \Rightarrow \tilde{z} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\text{Let } w = c + id \Rightarrow \tilde{w} = \begin{pmatrix} c \\ d \end{pmatrix}$$

$$\tilde{z} \cdot \tilde{w} = ac + bd$$

$$|\tilde{w}|^2 = c^2 + d^2$$

$$\text{proj}_{\tilde{w}} \tilde{z} = \frac{\tilde{z} \cdot \tilde{w}}{|\tilde{w}|^2} \cdot \tilde{w} = \frac{ac + bd}{c^2 + d^2} \cdot \tilde{w}$$

$$\begin{aligned} \frac{z}{w} &= \frac{a + ib}{c + id} \times \frac{c - id}{c - id} \\ &= \frac{ac + bd + i(bc - ad)}{c^2 + d^2} \end{aligned}$$

$$\text{Re}\left(\frac{z}{w}\right) = \frac{ac + bd}{c^2 + d^2}$$

$$\therefore \text{proj}_{\tilde{w}} \tilde{z} = \text{Re}\left(\frac{z}{w}\right) \tilde{w}$$

$$\Rightarrow C$$

◆◆◆ Mean mark 30%.

7. Vectors, EXT2 V1 2023 HSC 10 MC

By elimination

Option A:

If $\vec{a}$ and $\vec{b}$ are the two unit vectors, $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = \cos \theta$

$$-1 \leq \cos \theta \leq 1 \Rightarrow -1 \leq \vec{a} \cdot \vec{b} \leq 1$$

$$\text{Given } \vec{a} \cdot \vec{b} = 1 \Rightarrow \vec{a} = \vec{b} \Rightarrow \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a}$$

$$\text{Contradicts } \vec{b} \cdot \vec{c} = 2 \text{ and } \vec{c} \cdot \vec{a} = 3$$

Similar reasoning rules out any pair satisfying all conditions (eliminate A).

Option C: If $\vec{a} = \vec{0}$, $\vec{a} \cdot \vec{b} = 0 \neq 1$ (eliminate C).

Option D:

Consider the vectors below that satisfy the conditions,

$$\vec{a} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \vec{b} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{c} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

However, $r\vec{a} + s\vec{b} + t\vec{c} = \vec{0}$ requires $r = s = t = 0$ which are not positive constants (eliminate D).

$$\Rightarrow B$$

8. Vectors, EXT2 V1 EQ-Bank 2

$$\vec{r} = \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}, |\vec{r}| = \sqrt{3^2 + (-2)^2 + (-1)^2} = \sqrt{14}$$

$$\vec{s} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, |\vec{s}| = \sqrt{2^2 + 1^2 + 1^2} = \sqrt{6}$$

$$\vec{r} \cdot \vec{s} = \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 6 - 2 - 1 = 3$$

$$\vec{r} \cdot \vec{s} = |\vec{a}| |\vec{b}| \cos \theta$$

$$3 = \sqrt{14} \sqrt{6} \cos \theta$$

$$\cos \theta = \frac{3}{\sqrt{84}}$$

$$\theta = \cos^{-1} \left(\frac{3}{\sqrt{84}} \right)$$

$$= 70.9^\circ \text{ (1 d.p.)}$$

♦♦♦ Mean mark 15%.

9. Vectors, EXT2 V1 2021 HSC 11c

$$\vec{a} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix}, |\vec{a}| = \sqrt{2^2 + 4^2} = \sqrt{20}$$

$$\vec{b} = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}, |\vec{b}| = \sqrt{(-3)^2 + 1^2 + 2^2} = \sqrt{14}$$

$$\vec{a} \cdot \vec{b} = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix} = -6 + 0 + 8 = 2$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$2 = \sqrt{20} \sqrt{14} \cos \theta$$

$$\cos \theta = \frac{2}{\sqrt{280}}$$

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{70}} \right)$$

$$= 83.1^\circ \text{ (1 d.p.)}$$

10. Vectors, EXT2 V1 2023 HSC 11b

$$\vec{a} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}, \vec{b} = \begin{pmatrix} -1 \\ 4 \\ 2 \end{pmatrix}$$

$$|\vec{a}| = \sqrt{1 + 4 + 9} = \sqrt{14}$$

$$|\vec{b}| = \sqrt{1 + 16 + 4} = \sqrt{21}$$

$$\vec{a} \cdot \vec{b} = -1 + 8 - 6 = 1$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

$$= \frac{1}{\sqrt{294}}$$

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{294}} \right)$$

$$= 86.65 \dots$$

$$= 87^\circ$$

11. Vectors, EXT2 V1 EQ-Bank 1

$$\begin{aligned}\vec{a} \cdot \vec{b} &= \begin{pmatrix} 4 \\ 5 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -5 \\ 6 \\ 5 \end{pmatrix} \\ &= 4 \times (-5) + 5 \times 6 + (-2) \times 5 \\ &= -20 + 30 + 10 \\ &= 0\end{aligned}$$

Since $\vec{a} \cdot \vec{b} = 0 \Rightarrow \vec{a} \perp \vec{b}$

12. Vectors, EXT2 V1 EQ-Bank 3

$$\begin{aligned}\text{i. } \vec{a} - \frac{1}{2}\vec{b} &= \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -2 \\ 6 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ -4 \\ -1 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\text{ii. } \hat{\vec{b}} &= \frac{\vec{b}}{|\vec{b}|} \\ |\vec{b}| &= \sqrt{(-2)^2 + 6^2 + 2^2} = \sqrt{44} = 2\sqrt{11} \\ \hat{\vec{b}} &= \frac{1}{2\sqrt{11}}(-2, 6, 2) \\ &= \left(\frac{-1}{\sqrt{11}}, \frac{3}{\sqrt{11}}, \frac{1}{\sqrt{11}} \right) \\ &= \left(\frac{-\sqrt{11}}{11}, \frac{3\sqrt{11}}{11}, \frac{\sqrt{11}}{11} \right)\end{aligned}$$

13. Vectors, EXT2 V1 SM-Bank 5

$$\vec{a} \perp \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$$

$$\begin{aligned}\vec{a} \cdot \vec{b} &= 2m^2 + m(-1) + (-3)(1) \\ 0 &= 2m^2 - m - 3 \\ 0 &= (2m-3)(m+1)\end{aligned}$$

$$\therefore m = \frac{3}{2}, \quad m = -1$$

14. Vectors, EXT2 V1 SM-Bank 12

Using $|\vec{a}| = 10$:

$$10 = \sqrt{4^2 + m^2 + (-3)^2}$$

$$\begin{aligned}100 &= m^2 + 25 \\ m &= 5\sqrt{3}\end{aligned}$$

Since $\vec{a} \perp \vec{b}$:

$$\begin{aligned}\vec{a} \cdot \vec{b} &= 0 \\ 0 &= 4 \times (-2) + mn + (-3) \times (-1) \\ 0 &= mn - 5 \\ n &= \frac{5}{5\sqrt{3}} \\ &= \frac{\sqrt{3}}{3}\end{aligned}$$

15. Vectors, EXT2 V1 2020 HSC 11d

$$\vec{u} - \vec{v} = \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix} - \begin{pmatrix} p \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2-p \\ -2 \\ 1 \end{pmatrix}$$

$$\vec{u} + \vec{v} = \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix} + \begin{pmatrix} p \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} p-2 \\ 0 \\ 5 \end{pmatrix}$$

$$\perp \text{ when } (\vec{u} - \vec{v}) \cdot (\vec{u} + \vec{v}) = 0:$$

$$\begin{pmatrix} -2-p \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} p-2 \\ 0 \\ 5 \end{pmatrix} = 0$$

$$-(p+2)(p-2) + 5 = 0$$

$$-(p^2-4) + 5 = 0$$

$$-p^2 + 9 = 0$$

$$p^2 = 9$$

$$p = \pm 3$$

16. Vectors, EXT2 V1 2022 HSC 11d

$$\vec{BA} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix}$$

$$|\vec{BA}| = \sqrt{1^2 + 3^2 + 3^2} = \sqrt{19}$$

$$\vec{BC} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

$$|\vec{BC}| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$$

$$\vec{BA} \cdot \vec{BC} = 1 \times 2 + -3 \times -1 + 3 \times 2 = 11$$

$$\begin{aligned} \cos \angle ABC &= \frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| |\vec{BC}|} \\ &= \frac{11}{3\sqrt{19}} \end{aligned}$$

$$\therefore \angle ABC = \cos^{-1} \left(\frac{11}{3\sqrt{19}} \right)$$

$$= 32.733\dots$$

$$= 33^\circ \text{ (nearest degree)}$$

17. Vectors, EXT2 V1 2022 SPEC1 6

$$\begin{aligned} \cos \theta &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \\ &= \frac{2 - 6 + 12}{\sqrt{4 + 9 + 36} \sqrt{1 + 4 + 4}} \\ &= \frac{8}{7 \times 3} \\ &= \frac{8}{21} \end{aligned}$$

18. Vectors, EXT2 V1 EQ-Bank 12

Let $\vec{a} = -200\vec{i} + 400\vec{j} - 3\vec{k}$, $\vec{b} = 300\vec{i} + 250\vec{j} + \vec{k}$

$$\vec{a} \cdot \vec{b} = \begin{pmatrix} -200 \\ 400 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 300 \\ 250 \\ 1 \end{pmatrix} = -60\,000 + 100\,000 - 3 = 39\,997$$

$$|\vec{a}| = \sqrt{200^2 + 400^2 + 9} = 447.2236\dots$$

$$|\vec{b}| = \sqrt{300^2 + 250^2 + 1} = 390.5137\dots$$

$$\cos\theta = \frac{39\,997}{447.2236 \times 390.5137} \\ = 0.2291$$

$$\therefore \theta = 76.8^\circ \text{ (to 1 d.p.)}$$

19. Vectors, EXT2 V1 EQ-Bank 6

Method 1

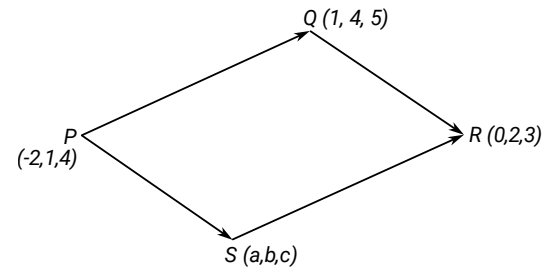
Let $\vec{c}$ = position vector of the dividing point

$$\vec{c} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 7 - (-1) \\ -1 - 3 \\ -6 - 2 \end{pmatrix} \\ = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} \\ = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

Method 2

$$\vec{c} = \begin{pmatrix} 7 \\ -1 \\ -6 \end{pmatrix} + \frac{3}{4} \begin{pmatrix} -1 - 7 \\ 3 - (-1) \\ 2 - (-6) \end{pmatrix} \\ = \begin{pmatrix} 7 \\ -1 \\ -6 \end{pmatrix} + \begin{pmatrix} -6 \\ 3 \\ 6 \end{pmatrix} \\ = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

20. Vectors, EXT2 V1 EQ-Bank 9



Opposite sides of a parallelogram are equal and parallel.

$$\Rightarrow \vec{PQ} = \vec{SR}$$

$$\vec{PQ} = \begin{pmatrix} 1 + 2 \\ 4 - 3 \\ 5 - 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$$

$$\vec{SR} = \begin{pmatrix} -a \\ 2 - b \\ 3 - c \end{pmatrix}$$

Equating coordinates:

$$-a = 3 \Rightarrow a = -3$$

$$2 - b = 1 \Rightarrow b = 1$$

$$3 - c = 1 \Rightarrow c = 2$$

21. Vectors, EXT2 V1 SM-Bank 11

$$\lambda_1 \vec{a} + \lambda_2 \vec{b} = \begin{bmatrix} 2\lambda_1 + 3\lambda_2 \\ -3\lambda_1 - 2\lambda_2 \\ 4\lambda_1 - 3\lambda_2 \end{bmatrix}$$

$$2\lambda_1 + 3\lambda_2 = 50 \dots (1)$$

$$4\lambda_1 - 3\lambda_2 = -8 \dots (2)$$

$$(1) + (2)$$

$$6\lambda_1 = 42$$

$$\lambda_1 = 7$$

Substitute $\lambda_1 = 7$ into (1):

$$14 + 3\lambda_2 = 50$$

$$\lambda_2 = 12$$

22. Vectors, EXT2 V1 SM-Bank 4

$$\begin{aligned} \text{i. } |\tilde{b}| &= \sqrt{1^2 + 2^2 + m^2} = 2\sqrt{3} \\ 1 + 4 + m^2 &= 4 \times 3 \\ m^2 &= 7 \\ m &= \pm \sqrt{7} \end{aligned}$$

$$\begin{aligned} \text{ii. } \tilde{a} \cdot \tilde{b} &= 1 \times 1 + (-1) \times 2 + 2 \times m \\ 0 &= 1 - 2 + 2m \quad (\tilde{a} \perp \tilde{b}) \\ 2m &= 1 \\ m &= \frac{1}{2} \end{aligned}$$

23. Vectors, EXT2 V1 2013 SPEC1 3

$$\begin{aligned} \text{i. } \overrightarrow{AB} &= (1-1)\tilde{i} + (0-2)\tilde{j} + (5-4)\tilde{k} \\ &= 2\tilde{i} - 2\tilde{j} + \tilde{k} \end{aligned}$$

$$\begin{aligned} \text{ii. } \overrightarrow{AC} &= (3-1)\tilde{i} + (5-2)\tilde{j} + (2-4)\tilde{k} \\ &= 4\tilde{i} + 3\tilde{j} - 2\tilde{k} \end{aligned}$$

$$\begin{aligned} \overrightarrow{AB} \cdot \overrightarrow{AC} &= 2 \times 4 + (-2) \times 3 + 1 \times (-2) \\ &= 8 - 6 - 2 \\ &= 0 \end{aligned}$$

$$\begin{aligned} &\Rightarrow \overrightarrow{AB} \perp \overrightarrow{AC} \\ &\therefore \Delta ABC \text{ has a right angle at } A. \end{aligned}$$

$$\begin{aligned} \text{iii. } \overrightarrow{BC} &= (3-1)\tilde{i} + (5-0)\tilde{j} + (2-5)\tilde{k} \\ &= 2\tilde{i} + 5\tilde{j} - 3\tilde{k} \end{aligned}$$

$$\begin{aligned} |\overrightarrow{BC}| &= \sqrt{2^2 + 5^2 + (-3)^2} \\ &= \sqrt{4 + 25 + 9} \\ &= \sqrt{38} \end{aligned}$$

24. Vectors, EXT2 V1 2014 SPEC1 1

$$\begin{aligned} \text{i. } |\tilde{a}| &= \sqrt{(\sqrt{3})^2 + (-1)^2 + (-\sqrt{2})^2} \\ &= \sqrt{6} \\ \therefore \hat{\tilde{a}} &= \frac{\tilde{a}}{|\tilde{a}|} \\ &= \frac{1}{\sqrt{6}} (\sqrt{3}\tilde{i} - \tilde{j} - \sqrt{2}\tilde{k}) \end{aligned}$$

$$\begin{aligned} \text{ii. } \tilde{a} \cdot \tilde{i} &= \sqrt{3} \times 1 = \sqrt{3} \\ \tilde{a} \cdot \tilde{i} &= |\tilde{a}| |\tilde{i}| \cos \theta \\ &= \sqrt{6} \cos \theta \\ \sqrt{3} &= \sqrt{6} \cos \theta \\ \cos \theta &= \frac{1}{\sqrt{2}} \\ \therefore \theta &= \frac{\pi}{4} = 45^\circ \end{aligned}$$

$$\begin{aligned} \text{iii. } \tilde{a} \cdot \tilde{b} &= \sqrt{3}(2\sqrt{3}) + (-1)(m) + (-\sqrt{2})(-5) = 0 \\ 6 - m + 5\sqrt{2} &= 0 \\ \therefore m &= 6 + 5\sqrt{2} \end{aligned}$$

Mean mark part (b) 51%.

25. Vectors, EXT2 V1 EQ-Bank 4

$$\overrightarrow{PQ} = \begin{pmatrix} -3 \\ -2 \\ 4 \end{pmatrix} - \begin{pmatrix} -5 \\ b \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -2-b \\ 6 \end{pmatrix}$$

$$|\overrightarrow{PQ}| = \sqrt{2^2 + (-2-b)^2 + 6^2}$$

$$76 = (-2-b)^2 + 40$$

$$36 = (-2-b)^2$$

$$6 = \pm (-2-b)$$

$$-2-b=6 \Rightarrow b=-8$$

$$2+b=6 \Rightarrow b=4$$

$$\therefore b=4 \text{ or } -8$$

26. Vectors, EXT2 V1 2017 SPEC1 10

$$\begin{aligned}\vec{a} \cdot \vec{b} &= -1 \times 2 + (-2) \times c + 3 \times 1 \\ &= -2 - 2c + 3 \\ &= 1 - 2c\end{aligned}$$

$$1 - 2c = \sqrt{(-1)^2 + (-2)^2 + 3^2} \cdot \sqrt{2^2 + c^2 + 1^2} \times \cos\left(\frac{\pi}{3}\right)$$

$$1 - 2c = \frac{1}{2} \left(\sqrt{14} \cdot \sqrt{5 + c^2} \right)$$

$$2 - 4c = \sqrt{14(5 + c^2)}$$

$$(2 - 4c)^2 = 14(5 + c^2)$$

$$4 - 16c + 16c^2 = 70 + 14c^2$$

$$2c^2 - 16c - 66 = 0$$

$$c^2 - 8c - 33 = 0$$

$$(c - 11)(c + 3) = 0$$

$$c = 11 \text{ or } c = -3$$

$$\text{Since } 2 - 4c = \sqrt{14(5 + c^2)}$$

$$2 - 4c > 0 \Rightarrow c < \frac{1}{2}$$

$$\therefore c = -3$$

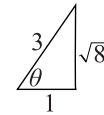
27. Vectors, EXT2 V1 SM-Bank 25

$$\vec{a} = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}, \quad |\vec{a}| = \sqrt{4 + 4 + 1} = 3$$

$$\vec{b} = \begin{pmatrix} -4 \\ 4 \\ 7 \end{pmatrix}, \quad |\vec{b}| = \sqrt{16 + 16 + 49} = 9$$

$$\vec{a} \cdot \vec{b} = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 4 \\ 7 \end{pmatrix} = 8 + 8 - 7 = 9$$

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{9}{3 \times 9} = \frac{1}{3}$$



$$\sin\theta = \frac{\sqrt{8}}{3}$$

$$\sin 2\theta = 2 \sin\theta \cos\theta$$

$$= 2 \cdot \frac{\sqrt{8}}{3} \cdot \frac{1}{3}$$

$$= \frac{4\sqrt{2}}{9}$$

28. Vectors, EXT2 V1 SM-Bank 7

$$\begin{aligned}\vec{a} \cdot \vec{b} &= \sqrt{3} \times 1 + 4 \times (-4) + (-1) \times \sqrt{3} \\ &= -16\end{aligned}$$

$$|\vec{a}| = \sqrt{20}, \quad |\vec{b}| = \sqrt{20},$$

$$\begin{aligned}\cos(\theta) &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \\ &= \frac{-16}{\sqrt{20} \times \sqrt{20}} \\ &= -\frac{4}{5}\end{aligned}$$

$$\begin{aligned}\cos(2\theta) &= 2\cos^2\theta - 1 \\ &= 2\left(-\frac{4}{5}\right)^2 - 1 \\ &= \frac{7}{25}\end{aligned}$$

29. Vectors, EXT2 V1 SM-Bank 8

$$\vec{a} - \vec{b} = (m-2)\vec{i} - 2\vec{j} + \vec{k}$$

$$(\vec{a} - \vec{b}) \cdot \vec{b} = -m(m-2) - 2(1) + 1(2) = 0$$

$$0 = -m^2 + 2m - 2 + 2$$

$$0 = 2m - m^2$$

$$0 = m(2-m)$$

$$\therefore m = 0 \text{ or } 2$$

30. Vectors, EXT2 V1 2020 SPEC1 5

$$\text{a. } \vec{b} \cdot \vec{a} = \begin{pmatrix} 1 \\ m \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = 2 - 3m - 1 = 1 - 3m$$

$$|\vec{b}|^2 = 1^2 + m^2 + (-1)^2 = m^2 + 2$$

$$\frac{\vec{b} \cdot \vec{a}}{|\vec{b}|^2} \cdot \vec{b} = -\frac{11}{18}\vec{b}$$

$$\frac{1 - 3m}{m^2 + 2} = -\frac{11}{18}$$

$$18 - 54m = -11m^2 - 22$$

$$11m^2 - 54m + 40 = 0$$

$$(11m - 10)(m - 4) = 0$$

$$\therefore m = 4 \quad \left(m \neq \frac{10}{11}, m \in \mathbb{Z}\right)$$

$$\begin{aligned}\text{b. } \vec{a} \perp \vec{b} &= \vec{a} + \frac{11}{18}(\vec{i} + 4\vec{j} - \vec{k}) \\ &= 2\vec{i} + \frac{11}{18}\vec{i} - 3\vec{j} + \frac{44}{18}\vec{j} + \vec{k} - \frac{11}{18}\vec{k} \\ &= \frac{47}{18}\vec{i} - \frac{5}{9}\vec{j} + \frac{7}{18}\vec{k}\end{aligned}$$

31. Vectors, EXT2 V1 2017 SPEC1 5

Angle between $\overrightarrow{CB}$ and $\overrightarrow{CD} = \frac{\pi}{3}$

$$\overrightarrow{CB} = (1-2)\hat{i} + (-1- -1)\hat{j} + (2-1)\hat{k}$$
$$= -\hat{i} + \hat{k}$$

$$\overrightarrow{CD} = (a-2)\hat{i} + (-2- -1)\hat{j} + (-1+0)\hat{k}$$
$$= (a-2)\hat{i} - \hat{j} - \hat{k}$$

$$\overrightarrow{CD} \cdot \overrightarrow{CB} = -(a-2) + 0 - 1$$
$$= 1-a$$
$$= \left| \overrightarrow{CD} \right| \left| \overrightarrow{CB} \right| \cos\left(\frac{\pi}{3}\right)$$

$$1-a = \sqrt{(a-2)^2 + (-1)^2 + (-1)^2} \sqrt{(-1)^2 + (1)^2} \cos\left(\frac{\pi}{3}\right)$$

$$1-a = \sqrt{a^2-4a+4+1+1} \times \sqrt{2} \times \frac{1}{2}$$

$$2(1-a) = \sqrt{2a^2-8a+12}, \quad a < 1$$

$$4(1-a)^2 = 2a^2-8a+12, \quad a < 1$$

$$4(1-2a+a^2) = 2a^2-8a+12$$

$$4-8a+4a^2 = 2a^2-8a+12$$

$$2a^2 = 8$$

$$a^2 = 4$$

$$\therefore a = -2 \quad (a < 1)$$

◆ Mean mark 45%.

32. Vectors, EXT2 V1 SM-Bank 9

$$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = -\hat{i} + 2\hat{j} - \hat{k}$$

$$\Rightarrow \left| \overrightarrow{BA} \right| = \sqrt{6}$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = -3\hat{i} - 2\hat{j}$$

$$\Rightarrow \left| \overrightarrow{BC} \right| = \sqrt{13}$$

$$\overrightarrow{BA} \cdot \overrightarrow{BC} = \left| \overrightarrow{BA} \right| \left| \overrightarrow{BC} \right| \cos \angle ABC$$

$$\cos \angle ABC = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{\left| \overrightarrow{BA} \right| \left| \overrightarrow{BC} \right|}$$
$$= \frac{-3 \times -1 + 2 \times -2}{\sqrt{6}\sqrt{13}}$$
$$= \frac{-1}{\sqrt{6}\sqrt{13}}$$

◆ Mean mark 48%.